

Sorghum (Jowar) Farming Guide for Pakistan

A complete crop guide for farmers, students, and agriculture advisors

1. Introduction

Sorghum, commonly known as Jowar or Sorghum-Chari in Pakistan, is a hardy cereal crop grown for grain, green fodder, and silage. It is one of the most important dryland cereals in the world and plays a similar role in Pakistan, particularly in areas with limited and unreliable rainfall. Sorghum is valued for its ability to withstand heat and drought better than maize, while still giving good yields of grain and especially fodder. It is grown both as a grain crop for human food and animal feed, and as a fodder crop (often called Chari) that is cut and fed green to livestock. Because of its dual-purpose nature and toughness, sorghum is an important part of mixed crop-livestock farming systems across Pakistan.

2. Scientific Name

The scientific name of sorghum is *Sorghum bicolor*, belonging to the family Poaceae. In Pakistan it is commonly called Jowar for the grain type and Chari for fodder-type sorghum, though both belong to the same species with different varieties suited to each purpose.

3. Importance in Pakistan

Sorghum holds significant importance in Pakistan's agricultural economy, especially as a fodder crop for the country's large livestock population. Chari sorghum is one of the most widely grown summer fodder crops, providing green feed for cattle and buffaloes during the hot months when other fodders are scarce. Grain sorghum is also used in poultry feed and, in some rural areas, as human food in the form of flour. Sorghum's drought tolerance makes it a reliable crop in rain-fed and water-short regions, helping farmers maintain livestock production even during dry years. It also fits well into crop rotations, helping to break pest and disease cycles associated with other cereals.

4. Major Producing Provinces and Districts

Sorghum is grown throughout Pakistan, with different districts favoring either grain or fodder types. - **Punjab:** Major fodder sorghum production is found in Sahiwal, Okara, Faisalabad, Multan, Bahawalpur, and Vehari, where it is grown mainly as summer fodder for the dairy industry. - **Sindh:** Grown widely in Hyderabad, Sanghar, Nawabshah (Shaheed Benazirabad), and Larkana, both for grain and fodder purposes. - **Balochistan:** Nasirabad, Jhal Magsi, and Sibi grow sorghum under both irrigated and rain-fed conditions. - **Khyber Pakhtunkhwa:** Grown on a smaller scale in Dera Ismail Khan, Bannu, and Peshawar valley, mainly as fodder. Punjab leads in fodder sorghum production due to its large dairy farming base.

5. Climate Requirements

Sorghum is a warm-season crop that thrives in temperatures between 26°C and 34°C and can tolerate short periods of extreme heat up to 40°C. It requires a frost-free growing period and is sensitive to cold temperatures, especially at germination and seedling stages. Sorghum is well known for its drought tolerance, thanks to its extensive root system and ability to become semi-dormant during moisture stress, resuming growth once water becomes available again. It performs well with moderate rainfall of around 400 to 600 millimeters but can survive on less, making it suitable for both irrigated and rain-fed farming across Pakistan.

6. Soil Requirements

Sorghum grows well in a wide range of soils, from sandy loam to clay loam, but it performs best in well-drained, fertile loamy soils. It can tolerate moderately saline and alkaline soils better than many other cereal crops, which is useful in parts of Sindh and southern Punjab where soil salinity is a concern. The ideal soil pH range is between 5.5 and 8.5. Sorghum does not tolerate waterlogging for extended periods, so fields should have proper drainage, especially during the monsoon season.

7. Land Preparation

Proper land preparation is essential for good seed germination and crop establishment. The field should be ploughed two to three times to obtain a fine, well-pulverized seedbed, followed by planking or leveling to ensure uniform water distribution during irrigation or rainfall. Old crop residues and weeds should be removed or incorporated during ploughing. In areas prone to waterlogging, raised beds or proper field channels should be prepared to allow excess water to drain away, since sorghum seedlings are sensitive to standing water.

8. Recommended Varieties (Pakistan)

Pakistan has several approved sorghum varieties for both grain and fodder purposes. - **Hegari:** A widely grown grain-type variety known for good yield potential. - **JS-2002:** A popular fodder sorghum variety used across Punjab and Sindh dairy belts. - **Chakwal Sorghum:** Suited for rain-fed areas of Punjab. - **Sudan Grass hybrids:** Sometimes grown in combination with sorghum for multi-cut fodder systems. - **Local Chari varieties:** Traditional fodder types still popular with smallholder livestock farmers. Farmers should choose varieties based on their main purpose — grain, single-cut fodder, or multi-cut fodder — and source certified seed from provincial seed corporations.

9. Seed Rate

Seed rate for sorghum depends on the intended use of the crop. - For grain production: 4 to 5 kilograms per acre when sown in rows. - For fodder (Chari) production: 25 to 30 kilograms per acre, since a dense stand encourages tender, leafy growth suitable for animal feed. - For multi-cut fodder systems: seed rate may be slightly higher, around 30 to 35 kilograms per acre, to support repeated cutting. Using the correct seed rate for the intended purpose ensures optimal plant density and quality.

10. Seed Treatment

Before sowing, sorghum seed should be treated with a fungicide such as Thiram or Carboxin to protect against seed-borne diseases and soil fungi that can cause poor germination. In regions with known termite problems, an insecticidal seed treatment is also recommended. Seed should be evenly coated with the treatment chemical and allowed to dry in shade before sowing, and treated seed should never be used for food or feed purposes.

11. Sowing Time (Province-wise)

- **Punjab:** Fodder sorghum is commonly sown from March to August in multiple sowings to ensure continuous fodder supply, while grain sorghum is typically sown from June to July.
- **Sindh:** Sowing generally takes place from March to August for fodder, and June to July for grain-type sorghum.
- **Balochistan:** Sowing is usually done from April to July, depending on local temperature and water availability.
- **Khyber Pakhtunkhwa:** Sorghum is sown mainly from June to July, coinciding with the monsoon season. Multiple staggered sowings are common for fodder sorghum to ensure a steady green fodder supply throughout summer.

12. Sowing Methods

Sorghum can be sown using different methods depending on the purpose. - **Broadcasting:** Common for fodder sorghum, where a dense, uniform stand is desired; seed is scattered evenly and then covered lightly with soil. - **Drilling in rows:** Preferred for grain sorghum, with row spacing of 45 to 60 centimeters, allowing better weeding, fertilizer placement, and airflow. - **Line sowing with seed drill:** Increasingly used on larger farms for efficiency and uniform plant stand. Grain sorghum benefits more from row sowing, while fodder sorghum is often broadcast for maximum biomass production.

13. Fertilizer Recommendations (NPK)

Balanced fertilization improves both grain and fodder yield of sorghum. - **Nitrogen (N):** 60 to 80 kilograms per acre for fodder sorghum (due to multiple cuttings), applied in split doses after each cutting; 50 to 60 kilograms per acre for grain sorghum, split between sowing and later growth stages. - **Phosphorus (P2O5):** 25 kilograms per acre, applied fully at sowing time. - **Potassium (K2O):** 15 to 20 kilograms per acre, particularly important for fodder sorghum to maintain plant vigor across multiple cuts. Farmers growing multi-cut fodder sorghum should apply a top dressing of nitrogen after each harvest to encourage vigorous regrowth.

14. Irrigation Schedule

Sorghum requires regular irrigation under irrigated farming systems, especially for fodder production where frequent cutting removes significant biomass. Typically, the first irrigation is given at sowing (or shortly after, if pre-sowing irrigation was not applied), followed by irrigations at approximately 10 to 15 day intervals depending on soil type and weather. For grain sorghum, the critical stages requiring adequate water are flowering and grain filling. For fodder sorghum with multiple cuts, an irrigation after each cutting helps ensure quick and vigorous regrowth. In rain-fed areas, sorghum relies on monsoon rains, and no additional irrigation is applied.

15. Weed Management

Weeds compete significantly with young sorghum plants, especially in the first four to five weeks after sowing. Effective control includes one or two hand weedings or hoeings, timed at around 20 and 40 days after sowing. Pre-emergence herbicides such as Atrazine can be used where large areas make manual weeding impractical, applied strictly according to label recommendations and local extension advice. For fodder sorghum with multiple cuts, weeding after each cut also helps maintain crop vigor and fodder quality.

16. Insect Pests

Common insect pests affecting sorghum in Pakistan include: - **Shoot fly:** Attacks young seedlings, causing deadheart symptoms similar to those in millet. - **Stem borer:** Larvae tunnel into stems, weakening plants and reducing yield. - **Sorghum midge:** Affects developing grains, reducing seed set. - **Aphids:** Can build up rapidly, especially in dry weather, sapping plant strength. - **Armyworm:** Occasionally causes damage to leaves, especially in fodder sorghum fields. Regular field scouting and timely application of recommended insecticides, only when pest populations reach damaging levels, help manage these pests effectively.

17. Diseases

Sorghum is affected by several diseases in Pakistan's growing conditions. - **Grain mold:** Affects maturing grain in humid conditions, reducing grain quality. - **Charcoal rot:** A stress-related fungal disease that becomes worse under drought conditions, causing stalk weakness and lodging. - **Leaf blight:** Causes brown lesions on leaves, reducing photosynthesis and fodder quality. - **Downy mildew:** Similar to the disease in millet, causing abnormal leafy growth in the earhead. Crop rotation, resistant varieties, and proper field sanitation help reduce disease incidence.

18. Deficiency Symptoms

Nutrient deficiency symptoms in sorghum can be identified through visual inspection. - **Nitrogen deficiency:** Yellowing of older leaves starting from the tip, along with stunted growth and thin stalks. - **Phosphorus deficiency:** Purplish discoloration of leaves and poor root and tiller development. - **Potassium deficiency:** Yellowing and scorching of leaf margins, weak stalks prone to lodging. - **Zinc deficiency:** Common in alkaline soils, appearing as pale interveinal stripes on younger leaves and stunted growth. Prompt correction through proper fertilizer application prevents yield losses caused by nutrient deficiencies.

19. Harvesting

For grain sorghum, harvesting is done when grains become hard and the earhead changes color from green to brown, usually 90 to 110 days after sowing depending on variety. Harvesting can be done manually with sickles, cutting the earheads or whole plants, followed by sun-drying and

threshing. For fodder sorghum, the first cutting is usually done 50 to 60 days after sowing when plants reach about 1 to 1.5 meters in height, and subsequent cuttings are taken every 30 to 40 days depending on regrowth and irrigation. Fodder should be harvested before flowering for the best nutritional quality and to avoid the risk of prussic acid (hydrocyanic acid) toxicity, which can develop in young or stressed sorghum plants.

20. Storage

Grain sorghum must be dried to about 12 percent moisture content before storage to prevent mold and insect damage. It should be stored in clean, dry, well-ventilated containers or bags, kept off the floor, and protected from moisture and pests. Fodder sorghum, when not fed fresh, can be preserved as silage by chopping and storing it in airtight pits or bunkers, which allows farmers to store excess green fodder for use during lean periods. Proper storage practices help maintain feed quality and reduce post-harvest losses for both grain and fodder.

21. Expected Yield in Pakistan

Average grain sorghum yield in Pakistan ranges from about 15 to 20 maunds per acre (560 to 750 kilograms per acre) under good management, while yields can reach up to 25 maunds per acre with improved varieties and irrigation. Fodder sorghum yield is much higher in terms of biomass, typically ranging from 300 to 500 maunds of green fodder per acre per cutting, with multiple cuttings possible in a single season. Actual yields depend heavily on variety, soil fertility, irrigation availability, and pest and disease management.

22. Government Recommendations

Provincial agriculture departments, including the Punjab Agriculture Department, Sindh Agriculture Extension, and research institutes such as the Fodder Research Institute Sargodha, provide regular advisories for sorghum growers. These recommendations typically emphasize using certified seed of approved varieties, following recommended sowing schedules to ensure staggered fodder availability, and applying balanced fertilizer based on soil testing where possible. Farmers are also advised to avoid feeding young or drought-stressed sorghum to livestock without proper care, due to the risk of prussic acid poisoning, and to consult local livestock and agriculture extension offices for the latest guidance.

23. Modern Farming Practices

Progressive dairy and crop farmers in Pakistan are adopting improved sorghum cultivation methods, including mechanized sowing with seed drills for uniform stands, use of multi-cut hybrid fodder varieties to maximize fodder yield per unit area, and mechanized fodder chopping and silage-making equipment to improve feed efficiency. Balanced fertilizer application based on soil testing, rather than fixed traditional doses, is also becoming more common among commercial dairy farms integrating sorghum into their feeding systems.

24. Precision Agriculture

Precision agriculture tools are being gradually introduced into sorghum farming in Pakistan, particularly on larger commercial dairy farms. These include soil moisture sensors to optimize irrigation scheduling for multi-cut fodder sorghum, GPS-guided machinery for precise row sowing and fertilizer placement, and satellite or drone-based monitoring to assess crop vigor and plan cutting schedules across large fodder blocks. While such technologies remain limited mostly to large commercial operations, awareness is growing among agricultural universities and extension services about their potential benefits for smallholder farmers as well.

25. Sustainable Farming

Sorghum is inherently well suited to sustainable farming due to its drought tolerance and relatively low input requirements compared to crops like maize. Sustainable practices being encouraged include crop rotation with legumes to restore soil nitrogen naturally, minimum tillage to conserve soil moisture and reduce erosion, and integrated pest management to reduce dependence on chemical pesticides. Silage-making from surplus fodder sorghum also reduces feed wastage and supports more resilient livestock feeding systems during dry periods, contributing to overall farm sustainability.

26. Summary Note

This guide reflects sorghum farming practices specific to Pakistan, covering key growing regions in Punjab, Sindh, Khyber Pakhtunkhwa, and Balochistan. Farmers are encouraged to verify current variety recommendations, fertilizer prices, and sowing schedules with their local agriculture extension office, since these details are updated periodically.